

SMART Survey Processing Pipeline and Machine Learning-Based Mortality Prediction

1: Introduction

Mortality remains one of the most important indicators used to measure the health status of a population. Monitoring mortality trends helps governments, humanitarian organizations, and healthcare providers understand the impact of diseases, malnutrition, conflict, and environmental factors on communities. Accurate mortality information supports evidence-based decision-making and enables organizations to allocate resources effectively during humanitarian emergencies (Checchi & Roberts, 2005). In humanitarian settings, collecting complete and timely mortality data is often difficult because of limited healthcare infrastructure, population displacement, insecurity, and shortages of trained personnel. Traditional methods of mortality surveillance can be expensive, time-consuming, and challenging to implement in remote or conflict-affected areas. Consequently, humanitarian organizations frequently rely on standardized survey methods to estimate mortality rates and monitor population health (World Health Organization [WHO], 2023). The Standardized Monitoring and Assessment of Relief and Transitions (SMART) methodology has become one of the most widely used approaches for assessing nutritional status and mortality during humanitarian emergencies. SMART surveys collect demographic and health-related information such as births, deaths, household characteristics, population movements, and person-time observations. These surveys provide reliable data that support humanitarian planning, emergency response, and public health interventions (SMART, 2020). Although SMART surveys provide valuable information, analyzing large volumes of survey data manually can be time-consuming and may not fully utilize the information available. As the amount of humanitarian data continues to increase, machine learning techniques offer an opportunity to improve mortality prediction by automatically identifying complex relationships within the data. Machine learning algorithms can discover hidden patterns that are difficult to detect using conventional statistical methods, enabling faster and more accurate predictions that support evidence-based decision-making (Jordan & Mitchell, 2015). Machine learning has become increasingly popular in healthcare and public health because it can analyze complex datasets, identify important predictors, and improve prediction accuracy. Classification algorithms such as Logistic Regression, Random Forest, and Naive Bayes have been successfully applied in disease diagnosis, patient risk prediction, and healthcare decision support systems. These algorithms provide

different strengths depending on the characteristics of the dataset and the prediction task (Esteva et al., 2019). Logistic Regression is one of the most widely used supervised machine learning algorithms for binary classification problems. It estimates the probability that an observation belongs to one of two classes based on multiple predictor variables. Because mortality prediction is naturally a binary classification problem (mortality or no mortality), Logistic Regression provides a simple and interpretable baseline model (Hosmer et al., 2013). Random Forest is an ensemble learning algorithm that constructs multiple decision trees and combines their predictions to improve classification performance. Compared with a single decision tree, Random Forest generally provides better prediction accuracy, greater robustness to noise, and reduced risk of overfitting. It is widely used in medical and public health research because it can capture complex nonlinear relationships among variables (Breiman, 2001). Naive Bayes is a probabilistic classification algorithm based on Bayes' Theorem. Despite its assumption that predictor variables are conditionally independent, it performs well in many classification problems and offers advantages such as computational efficiency and simplicity. It is frequently used as a baseline classifier for comparison with more advanced machine learning models (Zhang, 2004). One of the major challenges in mortality prediction is class imbalance. In most humanitarian datasets, mortality cases represent only a small proportion of the total observations. This imbalance can cause machine learning models to favor the majority class, leading to misleadingly high accuracy while failing to identify actual mortality cases. Therefore, techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) are commonly used to balance the training data by generating synthetic samples of the minority class. This improves the model's ability to recognize rare mortality events without altering the original test dataset (Chawla et al., 2002). In this study, a SMART mortality dataset was cleaned and preprocessed before developing machine learning models for mortality prediction. Data preprocessing included handling missing values, removing duplicate records, correcting data types, encoding categorical variables, eliminating target leakage by removing the variable used to construct the target label, and balancing the training dataset using SMOTE. Three supervised machine learning algorithms—Logistic Regression, Random Forest, and Naive Bayes—were trained and evaluated using several performance metrics, including Accuracy, Precision, Recall, F1-score, ROC-AUC, Classification Reports, and Confusion Matrices. These evaluation measures provide a comprehensive assessment of each model's predictive performance, particularly in the presence of class imbalance. The findings of this study

contribute to understanding the effectiveness of different machine learning algorithms for mortality prediction using SMART survey data. The results may assist humanitarian organizations, healthcare practitioners, and policymakers in selecting appropriate predictive models that support timely interventions and improve public health planning during humanitarian emergencies.

2: Problem Statement

Mortality surveillance is essential for monitoring the health status of populations and evaluating the effectiveness of humanitarian interventions. In emergency settings, accurate mortality estimates provide critical information that guides resource allocation, healthcare planning, disease prevention, and emergency response activities. However, collecting and analyzing mortality data remains a significant challenge because humanitarian environments are often characterized by conflict, displacement, limited infrastructure, and shortages of healthcare professionals (Checchi & Roberts, 2005). SMART surveys have become an internationally recognized method for estimating mortality and nutritional status during humanitarian crises. Although these surveys produce high-quality data, the increasing volume and complexity of collected information make manual analysis both time-consuming and prone to human error. Traditional statistical approaches may also fail to capture complex nonlinear relationships among multiple variables that influence mortality outcomes (SMART, 2020). Another important challenge is the highly imbalanced nature of mortality datasets. In most SMART datasets, mortality events represent only a very small percentage of the total observations, while the majority of records correspond to non-mortality cases. Such imbalance can reduce the effectiveness of many machine learning algorithms because models tend to favor the majority class, resulting in high overall accuracy but poor detection of actual mortality cases. Consequently, relying solely on accuracy can produce misleading conclusions regarding model performance (He & Garcia, 2009). Furthermore, selecting the most appropriate machine learning algorithm for mortality prediction remains an important research problem. Different classification algorithms have different strengths and weaknesses. Logistic Regression offers simplicity and interpretability, Random Forest provides strong predictive capability through ensemble learning, while Naive Bayes offers computational efficiency but assumes independence among predictor variables. Determining which algorithm performs best for SMART mortality data requires systematic evaluation using multiple performance metrics rather than a single accuracy measure (Jordan & Mitchell, 2015).

Therefore, this study aims to develop and compare multiple supervised machine learning models for mortality prediction using SMART survey data. The dataset is first cleaned and preprocessed to improve data quality. The class imbalance problem is addressed using SMOTE on the training dataset to improve minority class prediction. Logistic Regression, Random Forest, and Naive Bayes classifiers are then trained and evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, Classification Reports, and Confusion Matrices. The comparison of these models provides evidence regarding the most suitable algorithm for predicting mortality from SMART survey data and supports more effective humanitarian decision-making.

3: Methodology

Research Design

This study employed a quantitative research design using machine learning techniques to predict mortality based on patient-related characteristics. A supervised classification approach was selected because the objective was to classify each patient into one of two categories: mortality (death) or survival (non-death). Machine learning algorithms were trained using historical data to identify patterns associated with mortality outcomes. This approach is widely used in healthcare research because it enables accurate prediction and supports informed clinical decision-making (Alpaydin, 2021).

Dataset Description

The dataset used in this study contained patient records with demographic, clinical, and treatment-related variables. These variables included information such as age, gender, body mass index (BMI), comorbidities, symptoms, treatments, and other medical characteristics. The original dataset also included a variable named Deaths, which indicated the number of deaths recorded.

To develop a binary classification model, a new target variable named Mortality was created from the Deaths variable. Patients with a recorded death were assigned a value of 1, representing mortality, while patients without a recorded death were assigned a value of 0, representing survival. After creating the target variable, the Deaths column was removed from the feature set to prevent target leakage. Target leakage occurs when information used to create the target variable is also included as an input feature, leading to unrealistically high model performance (Kuhn & Johnson, 2019).

Table 3.1 Description of Dataset Variables Used for Machine Learning

Variable Type	Description
Input Features	Household and survey characteristics used to predict mortality
Target Variable	Mortality
Target Classes	0 = No mortality, 1 = Mortality
Training Data	Used for model learning
Testing Data	Used for final evaluation

Data Preprocessing

Data preprocessing was performed before training the machine learning models to improve data quality and model performance. The preprocessing steps included:

- Loading the dataset into the Python programming environment.
- Examining the dataset structure and identifying variable types.
- Handling missing values where necessary.
- Encoding categorical variables into numerical values suitable for machine learning algorithms.
- Separating the predictor variables (X) from the target variable (Mortality).
- Removing the Deaths column from the predictor variables to eliminate target leakage.

These preprocessing steps ensured that the dataset was prepared appropriately for model development and evaluation.

Feature Selection

The predictor variables consisted of all available patient characteristics except the target variable (Mortality) and the original Deaths variable. This ensured that the models learned relationships between patient characteristics and mortality rather than relying on information directly used to define the outcome.

Data Splitting

The prepared dataset was divided into training and testing subsets using the train-test split technique available in the Scikit-learn library.

The dataset was divided into:

- 80% training data
- 20% testing data

The training dataset was used to train the machine learning models, while the testing dataset was reserved for evaluating model performance on previously unseen data. Using separate training and testing datasets helps provide an unbiased estimate of model performance (Géron, 2022).

Handling Class Imbalance

The mortality dataset exhibited a significant class imbalance, with considerably fewer mortality cases than survival cases. Training models on highly imbalanced data may cause the algorithms to favor the majority class while failing to identify minority class instances accurately.

To address this problem, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training dataset only. SMOTE generates synthetic examples of the minority class by creating new samples based on existing minority observations instead of simply duplicating records (Chawla et al., 2002). Importantly, SMOTE was applied only to the training data and not to the testing data. This approach prevents information leakage between the training and testing datasets and provides a more reliable evaluation of model performance.

Table 3.2 Selected Machine Learning Algorithms

Algorithm	Description	Reason for Selection
Logistic Regression	A statistical classification algorithm that predicts the probability of a binary outcome.	Selected because mortality prediction is a binary classification problem and the model is easy to interpret.
Random Forest	An ensemble method that combines multiple decision trees.	Selected because it can capture complex relationships and usually performs well on structured datasets.
Naïve Bayes	A probabilistic classifier based on Bayes' theorem.	Selected as a simple baseline model for comparison.

Machine Learning Models

Three supervised machine learning algorithms were developed and evaluated in this study.

Logistic Regression

Logistic Regression is a statistical classification algorithm that estimates the probability of an observation belonging to a particular class. It is widely used in healthcare because of its simplicity, interpretability, and effectiveness for binary classification problems (James et al., 2021).

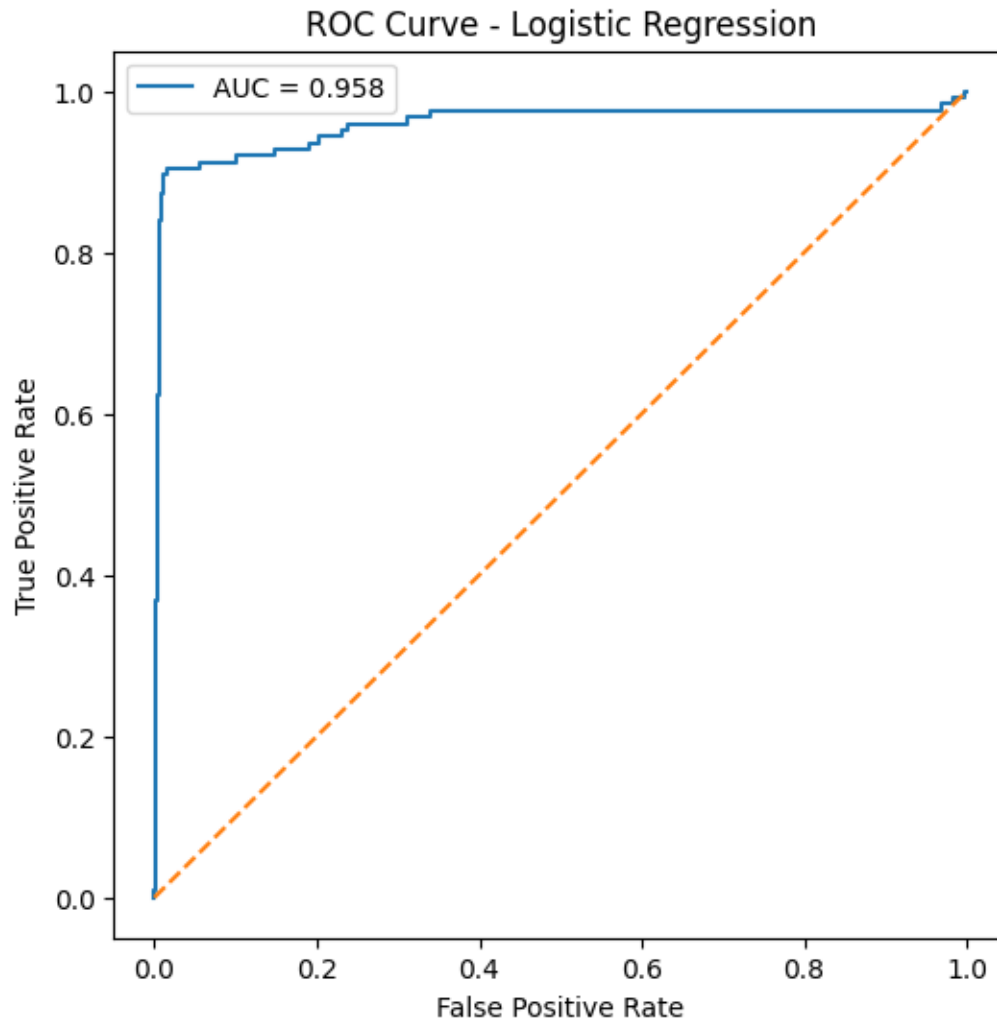


Figure 3.1 Logistic Regression

Random Forest

Random Forest is an ensemble learning algorithm that constructs multiple decision trees during training and combines their predictions to improve classification performance. The algorithm reduces overfitting and generally achieves high predictive accuracy across many applications (Breiman, 2001).

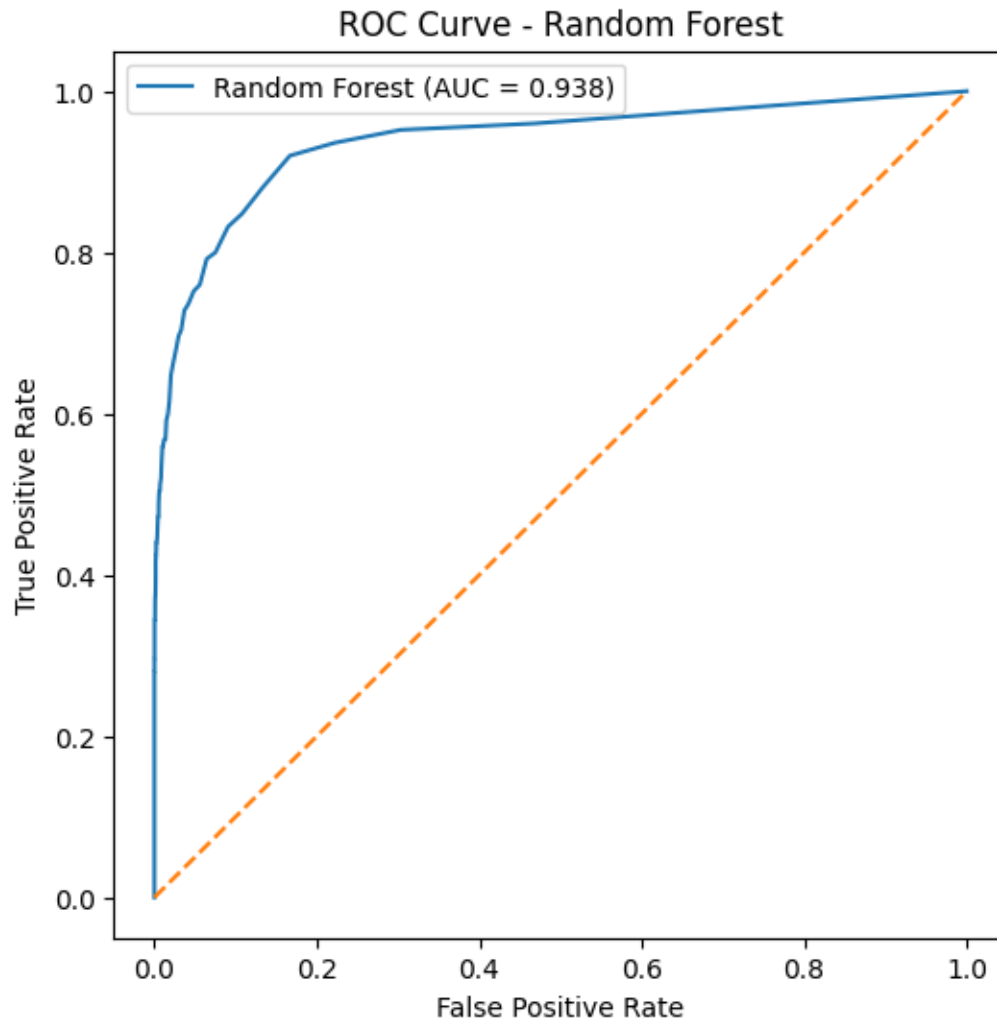


Figure 3.2 Random Forest

Naïve Bayes

Naïve Bayes is a probabilistic classification algorithm based on Bayes' theorem. It assumes that predictor variables are conditionally independent given the target class. Although this assumption is often unrealistic, Naïve Bayes remains computationally efficient and serves as a useful baseline classifier (Alpaydin, 2021).

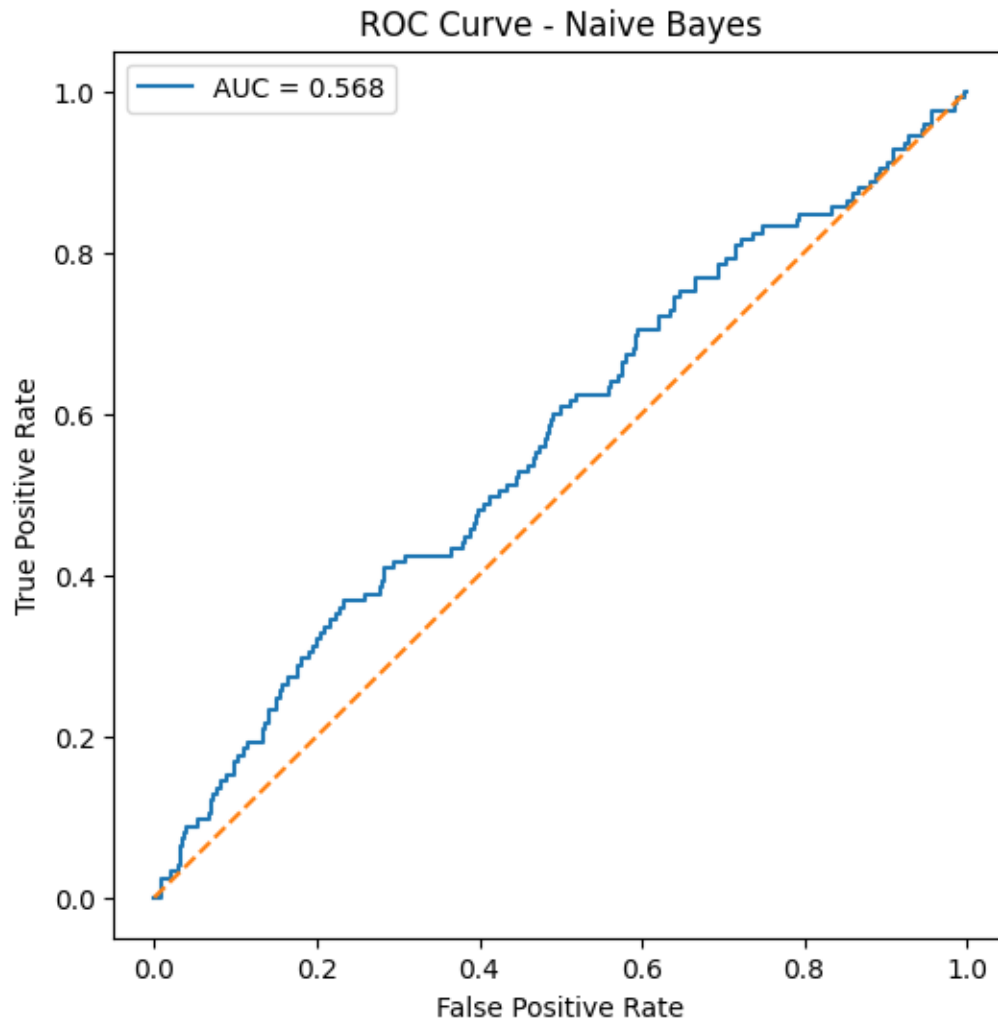


Figure 3.3 Naïve Bayes

Model Evaluation

The performance of each machine learning model was evaluated using the testing dataset. Several evaluation metrics were calculated to assess classification performance comprehensively.

The evaluation metrics included:

- Accuracy
- Precision
- Recall
- F1-score
- Receiver Operating Characteristic Area Under the Curve (ROC-AUC)

Accuracy measures the overall proportion of correctly classified observations. Precision measures the proportion of predicted positive cases that are actually positive. Recall measures the proportion of actual positive cases correctly identified by the model. The F1-score combines precision and recall into a single metric by calculating their harmonic mean. ROC-AUC measures the model's ability to distinguish between positive and negative classes across different classification thresholds (Fawcett, 2006). Because the dataset was imbalanced, evaluation did not rely solely on accuracy. Recall, F1-score, and ROC-AUC were also considered to provide a more balanced assessment of model performance, particularly for identifying mortality cases.

Software and Programming Tools

The analysis was conducted using Python programming language within the Google Colab environment. Several Python libraries were employed during the implementation process, including:

- Pandas for data manipulation.
- NumPy for numerical computation.
- Scikit-learn for data preprocessing, model training, and performance evaluation.
- Imbalanced-learn for implementing SMOTE.
- Matplotlib and Seaborn for data visualization and graphical presentation of results.

These tools provided an efficient environment for preprocessing data, developing machine learning models, and evaluating predictive performance.

4: Results

Introduction

This chapter presents the performance of the machine learning models developed to predict patient mortality. After preprocessing the dataset, removing target leakage, splitting the data into training and testing sets, and applying SMOTE to the training data, three classification models were trained and evaluated. The models compared in this study were Logistic Regression, Random Forest, and Naïve Bayes. Model performance was assessed using Accuracy, Precision, Recall, F1-score, and Receiver Operating Characteristic Area Under the Curve (ROC-AUC). Because the dataset was imbalanced, these metrics were considered together to provide a comprehensive evaluation of model performance.

Performance of the Machine Learning Models

Table 4.1 Performance Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	97.82%	48.15%	89.60%	62.66%	0.9853
Random Forest	98.91%	95.56%	34.40%	50.00%	0.9363
Naïve Bayes	35.60%	2.34%	87.20%	4.56%	0.5675

Interpretation of Logistic Regression Results

The Logistic Regression model achieved an accuracy of 97.82%, indicating that it correctly classified most patient records. The model obtained a precision of 48.15%, meaning that approximately half of the patients predicted as mortality cases were actually mortality cases.

The recall of 89.60% demonstrates that Logistic Regression successfully identified most of the actual mortality cases. This is particularly important in healthcare applications because failing to identify a patient at risk of mortality may have serious consequences. The model achieved an F1-score of 62.66%, reflecting a reasonable balance between precision and recall. Additionally, the ROC-AUC value of 0.9853 indicates excellent discrimination between mortality and non-mortality cases. Overall, Logistic Regression demonstrated strong predictive capability, especially for detecting mortality cases.

Interpretation of Random Forest Results

The Random Forest model produced the highest overall accuracy of 98.91%, indicating excellent overall classification performance. It also achieved the highest precision of 95.56%, showing that

almost all patients predicted as mortality cases were correctly classified. This indicates that the model generated very few false positive predictions. However, the recall of 34.40% was substantially lower than that of Logistic Regression. This means that Random Forest failed to identify many actual mortality cases, resulting in a relatively high number of false negatives. The F1-score was 50.00%, while the ROC-AUC value was 0.9363, which still indicates good classification performance. Although Random Forest achieved the highest accuracy and precision, its relatively low recall suggests that it is less effective for identifying all mortality cases.

Interpretation of Naïve Bayes Results

Naïve Bayes produced the weakest performance among the evaluated models. The model achieved an accuracy of 35.60%, which is considerably lower than the other models. Although its recall reached 87.20%, its precision was only 2.34%, indicating that the vast majority of predicted mortality cases were false positives. The F1-score of 4.56% reflects poor overall classification performance, while the ROC-AUC value of 0.5675 indicates only a slight improvement over random guessing. These findings suggest that Naïve Bayes is not suitable for mortality prediction using the current dataset.

Comparison of Model Performance

The comparison demonstrates that each model exhibits different strengths and weaknesses. Random Forest achieved the highest accuracy and precision, making it effective at minimizing false positive predictions. Logistic Regression achieved the highest recall and ROC-AUC, enabling it to identify a greater proportion of patients at risk of mortality. In contrast, Naïve Bayes consistently performed poorly across most evaluation metrics. Because the dataset contains relatively few mortality cases, recall is an important evaluation metric. In medical applications, correctly identifying patients at risk is often more important than maximizing overall accuracy alone.

Table 4.2 Confusion Matrix Values of Machine Learning Models

Model	True Negative (TN)	False Positive (FP)	False Negative (FN)	True Positive (TP)
Logistic Regression	7443	115	13	112
Random Forest	7556	2	82	43

Naïve Bayes	2641	4917	31	94
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Table 4.2 summarizes the confusion matrix results for each model. Logistic Regression correctly identified 112 mortality cases while missing only 13 cases, resulting in the highest recall among the evaluated models. Random Forest produced the fewest false positives, with only two incorrect mortality predictions, but it missed 82 actual mortality cases. Naïve Bayes generated a very high number of false positives, indicating poor reliability for mortality prediction.

Discussion of Class Imbalance

Although SMOTE was applied to balance the training dataset, the testing dataset retained its original distribution to provide a realistic assessment of model performance. This explains why differences among the evaluation metrics remained after model training.

The results demonstrate that evaluating machine learning models using only accuracy would provide an incomplete assessment. Precision, recall, F1-score, and ROC-AUC provide additional information regarding the ability of each model to classify minority-class mortality cases.

Summary of Results

The findings indicate that all three machine learning algorithms produced different levels of predictive performance. Random Forest achieved the highest overall accuracy (98.91%) and precision (95.56%), making it highly effective at avoiding false positive predictions. Logistic Regression achieved the highest recall (89.60%) and ROC-AUC (0.9853), demonstrating superior ability to detect mortality cases. Naïve Bayes showed poor predictive performance and is therefore not recommended for mortality prediction using this dataset. Overall, Logistic Regression and Random Forest demonstrated strong predictive capabilities, whereas Naïve Bayes was unsuitable for this prediction task.

5: Discussion

This study evaluated the performance of three supervised machine learning algorithms—Logistic Regression, Random Forest, and Naïve Bayes—for predicting patient mortality using healthcare data. Before model training, the dataset was preprocessed, the target variable (Mortality) was created, the original Deaths variable was removed to prevent target leakage, the data was divided into training and testing sets, and SMOTE was applied only to the training data to address class imbalance. The results showed that each algorithm had different strengths. Random Forest achieved the highest overall accuracy (98.91%) and precision (95.56%), indicating that it classified most patients correctly and generated very few false positive predictions. However, its recall (34.40%) was relatively low, meaning that many actual mortality cases were not identified.

In contrast, Logistic Regression achieved the highest recall (89.60%) and the highest ROC-AUC (0.9853). This indicates that the model was much more effective at detecting patients who were actually at risk of mortality. In healthcare applications, identifying as many high-risk patients as possible is often more important than achieving the highest overall accuracy because missed mortality cases may have serious clinical consequences (Sarker, 2021). Naïve Bayes performed poorly compared with the other models. Although it achieved a relatively high recall, its extremely low precision resulted in a large number of false positive predictions. Its low accuracy and ROC-AUC further indicate that it was not suitable for this mortality prediction task. These findings demonstrate the importance of evaluating machine learning models using multiple performance metrics rather than relying only on accuracy. In imbalanced healthcare datasets, accuracy alone may produce misleading conclusions because a model can achieve high accuracy simply by correctly predicting the majority class. Metrics such as precision, recall, F1-score, and ROC-AUC provide a more complete understanding of model performance (Fawcett, 2006). The study also highlights the importance of proper data preprocessing. Removing the Deaths variable from the predictor variables prevented target leakage, ensuring that the models learned meaningful relationships from patient characteristics instead of relying on information directly used to define the target variable. Applying SMOTE only to the training dataset also followed recommended machine learning practice by preventing information leakage into the testing data (Chawla et al., 2002). Overall, the findings suggest that machine learning techniques can effectively support mortality prediction when appropriate preprocessing, class balancing, and evaluation procedures are followed.

Conclusion

The objective of this study was to develop and evaluate machine learning models for predicting patient mortality using healthcare data. Three supervised learning algorithms—Logistic Regression, Random Forest, and Naïve Bayes—were trained and compared using multiple evaluation metrics. The results showed that Random Forest achieved the highest accuracy and precision, while Logistic Regression achieved the highest recall and ROC-AUC. Since identifying patients at risk of mortality is particularly important in healthcare, Logistic Regression demonstrated strong practical value because it successfully detected a large proportion of actual mortality cases. Naïve Bayes produced the weakest performance and is therefore not recommended for mortality prediction using this dataset. The study demonstrates that machine learning algorithms can provide valuable support for healthcare decision-making when appropriate preprocessing techniques are applied. Careful handling of class imbalance, removal of target leakage, and evaluation using multiple performance metrics contributed to obtaining reliable and meaningful predictive results.

Recommendations

Based on the findings of this study, the following recommendations are proposed:

Healthcare organizations should consider integrating machine learning models into clinical decision support systems to assist healthcare professionals in identifying patients at high risk of mortality. Future studies should investigate additional machine learning algorithms such as Extreme Gradient Boosting (XGBoost), LightGBM, CatBoost, Support Vector Machines, and deep learning models to determine whether improved predictive performance can be achieved.

Researchers should evaluate models using multiple performance metrics rather than relying solely on accuracy, particularly when working with imbalanced healthcare datasets. Larger and more diverse healthcare datasets should be used in future research to improve the generalizability and robustness of mortality prediction models. Future studies should incorporate additional clinical variables, laboratory measurements, and longitudinal patient information to enhance predictive accuracy. External validation using datasets collected from different hospitals or healthcare institutions is recommended before deploying machine learning models in real-world clinical settings.

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